

GYNOFLOR® vaginal tablets

NAME OF THE MEDICINAL PRODUCT

Gynoflor vaginal tablets

QUALITATIVE AND QUANTITATIVE COMPOSITION

Each vaginal tablet contains 100 million viable *Lactobacillus acidophilus* and 0.03 mg Estriol
For excipients, see “List of Excipients”.

PHARMACEUTICAL FORM

Vaginal tablets. The vaginal tablets are slightly beige, spotted, oval and biconvex.

CLINICAL PARTICULARS

Therapeutic indications

- restoration of the Lactobacillus flora after local and/or systemic treatment with anti-infective or chemotherapeutic agents;
- atrophic vaginitis due to estrogen deficiency during menopause and post-menopause, or as comedication to systemic hormone replacement therapy;
- vaginal discharge of unknown origin or mild to intermediate cases of bacterial vaginosis and candidiasis, where use of anti-infective therapy is not absolutely necessary.

A further possible application is the prophylactic treatment of recidive vaginal infections.

Posology and method of administration

Restoration therapy, vaginal discharge of unknown origin, mild to intermediate vaginal infections:

1 – 2 vaginal tablets daily for 6 – 12 days.

The vaginal tablets should be inserted deeply into the vagina in the evenings before retiring. This is best performed in a reclining position with the legs slightly bent.

Treatment should be interrupted during menstruation and resumed afterwards.

Atrophic vaginitis:

In cases of atrophic vaginitis, a dose of 1 vaginal tablet daily is recommended for 6 – 12 days, followed by a maintenance dose of 1 vaginal tablet at 1 – 2 days per week.

Instructions for use / handling

Gynoflor contains excipients which do not dissolve completely, such that remains of the tablet could occasionally be found in the panties. This is of no importance for the efficacy of Gynoflor.

In rare cases of a very dry vagina, it is possible that the vaginal tablet does not dissolve and is discharged by the vagina as intact tablet. As consequence, the treatment is not optimal. However, this is not harmful to the vagina. For prevention, the vaginal tablet can be moistened with a drop of water before insertion into a very dry vagina.

Patients should use a sanitary towel or panty liner. Patients should not use vaginal douches or rinses during treatment with Gynoflor.

Contraindications

- Hypersensitivity to the active substances or to any of the excipients
- Malignant changes (estrogen-dependent tumours) in the breast, uterus or vagina
- Endometriosis (suspected or manifest)
- Vaginal haemorrhaging of unknown origin
- Young girls who have not reached sexual maturity should not use this preparation

Special warnings and special precautions for use:

Read carefully the instructions before use.

Consult your doctor for further information.

Use upon doctor's prescription only.

Interaction with other medicaments and other forms of interaction

Lactobacillus acidophilus is sensitive to various anti-infective agents (local or systemic). Simultaneous treatment of such agents may lead to a reduction in the efficacy of Gynoflor.

Patient should inform doctor or pharmacist if taking or has recently taken any other medicines, even those not prescribed.

Pregnancy and lactation

Gynoflor can be used during pregnancy and lactation. However, as with medicinal products in general, caution should be exercised when prescribing Gynoflor to pregnant women in the first trimester. Data on a limited number (54) of exposed pregnancies indicate no adverse effects of estriol and *Lactobacillus acidophilus* on pregnancy or on the health of the foetus/new-born child. Furthermore, Gynoflor has been used for more than 15 years without apparent adverse effects.

Animal studies have shown harmful effects of estriol on male foetuses (see “Preclinical safety data”). However, estriol is a sexual hormone which is specific for humans and is not found in other mammalian species. Thus, toxicological data from animal studies are of limited use and can not be extrapolated to humans.

Gynoflor contains estriol only in a very low dose. It has been shown in non-pregnant women that after the first application of Gynoflor, the plasma concentration of estriol increases transiently, whereas after the 12th application no increase was observed. After repeated application, the plasma concentration of estriol is within the normal range of post-menopausal women.

Furthermore, the maternal plasma concentration of estriol during pregnancy is increased about 1000 times as compared to non-pregnant women.

Effects on ability to drive and use machines

Not relevant

Undesirable effects

General disorders and administration site conditions

A mild stinging or burning (1.6 %) may be experienced shortly after the administration of Gynoflor. In rare cases, intolerance reactions such as redness and itching have been reported. In one case, an allergy to the Lactobacillus lyophilisate contained in Gynoflor was observed.

If Gynoflor is accidentally administered orally, no adverse effects are to be expected.

Patient should be advised to inform doctor or pharmacist of any side effects related to the drug use.

Overdose

No case of overdose has been reported.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: Gynaecological anti-infective and antiseptic

ATC code: G01AX99

Lactobacillus acidophilus is one of the predominant microorganisms of the vaginal flora in healthy women. Lactobacilli are non-pathogenic bacteria and have a protective function in the vagina. They ferment glycogen stored in the vaginal epithelium to lactic acid. The resulting acidic environment (pH 3.8 - 4.5) provides unfavourable conditions for the colonisation and growth of pathogenic microorganisms and provides an optimal medium for the proliferation of the lactobacilli.

In addition to lactic acid, lactobacilli produce hydrogen peroxide and bacteriocins, which also inhibit the growth of pathogenic microorganisms.

Estriol is an endogenous estrogenic hormone which has a specific effect on the vagina, but not on the endometrium. The purpose of the estriol in Gynoflor is to ensure proliferation and maturation of the vaginal epithelium. A proliferated and matured vaginal epithelium acts as physical barrier and stores glycogen, a nutritional substrate for lactobacilli. In the case of hormonal disorders, more commonly observed at advanced age, the vaginal epithelium is disturbed and the glycogen content decreased. Vaginal infections can also lead to a damaged vaginal epithelium.

Exogenous estriol, even in the very low dose contained in Gynoflor, improves the proliferation and maturation of the epithelium, and thus provides the optimal conditions for the restoration of the vaginal lactobacilli flora.

Disturbance or elimination of the physiological vaginal flora can occur mainly as a result of local or systemic treatment with anti-infective agents, serious general illnesses, improper hygienic measures and vaginal infections. In a non-physiological vaginal environment, the lactobacilli are reduced in numbers or even absent, and their protective function is no longer assured.

The therapeutic concept of Gynoflor is based on the restoration of the vaginal flora through exogenous lactobacilli and the improved proliferation and maturation of the vaginal epithelium by exogenous estriol, resulting in the re-establishment of the physiological equilibrium of the vaginal flora and the vaginal epithelium.

Lactobacillus acidophilus and estriol exert their actions locally within the vagina.

Lactose, used for the manufacturing of the vaginal tablets, can also be fermented to lactic acid by the lactobacilli. Reproduction of the lactobacilli and recolonisation of the vagina by these bacteria does start after the first application of the preparation.

Pharmacokinetic properties

As soon as the vaginal tablet comes into contact with vaginal secretion, the tablet begins to disintegrate, and lactobacilli as well as estriol are released. *In vitro* experiments have demonstrated that the lactobacilli resume their metabolism and cause a reduction in pH within a few hours.

The positive estrogenic effect induced by estriol is also quickly initiated, and the proliferation and maturation status of the vaginal epithelium improves progressively over the course of treatment (6 – 12 days).

The absorption of estriol from Gynoflor was investigated in healthy, post-menopausal women with atrophic epithelia. After a single, intravaginal application of Gynoflor, there was an increase in the plasma concentration of unconjugated estriol (biologically active form) above the basal level, and the maximum estriol concentration was reached after 3 hours. After 8 hours, the plasma concentration of unconjugated estriol was no longer elevated. However after the 12th application at the end of the therapy with Gynoflor® (once daily), the peak plasma level of unconjugated estriol was not elevated as compared to the basal level.

During the 12-day treatment, no accumulation of estriol was observed. The plasma concentrations observed after repeated application of Gynoflor were within the normal range of the endogenous, post-menopausal plasma level of unconjugated estriol.

The plasma concentrations of estrone and estradiol are not affected by estriol therapy, since estriol is their metabolic end-product. Estriol is eliminated

rapidly in the urine, primarily in biologically inactive forms (glucuronides, sulphates).

Due to the low dose of estriol (0.03 mg), the local application and the limited duration of treatment (6 – 12 days), no accumulation of estriol occurs and systemic estrogen effects are very unlikely.

Preclinical safety data

Two toxicological aspects must be considered for drugs administered by the vaginal route: local tolerance and the possibility of systemic toxicity due to absorption of the components.

Lactobacilli are the predominant microorganisms in the healthy human vagina and are regarded as non-pathogenic. For this reason, the introduction of these bacteria into the vagina is not expected to cause any local irritation or epithelial damage.

Results of animal experiments (Wistar rats) have demonstrated that only high doses of estriol can lead to a feminisation of the male foetus.

It should be noted that toxicological data obtained in rodents are of limited value and can not be extrapolated to man, since estriol is specific for humans and does not occur in other mammalian species. Considering the low dose of estriol in Gynoflor, no acute or chronic toxicity is to be expected.

PHARMACEUTICAL PARTICULARS

List of excipients

Lactose, Disodium phosphate anhydrous, Cellulose microcrystalline, Sodium starch glycolate, Magnesium stearate

Incompatibilities

Not applicable

Shelf-life

36 months from the manufacturing date

Specification

Manufacturer

Special precautions for storage

Store in a refrigerator (2 °C – 8 °C).

Storage of Gynoflor at room temperature during the 1 – 2 weeks treatment period does not affect its efficacy.

Keep medicines out of the reach of children.

Do not use the medicine after the expiry date stated on the package.

Nature and contents of container

PVC/aluminium blisters

Presentations: Box of 1 blister of 6 vaginal tablets.
Box of 2 blisters of 6 vaginal tablets.

DATE OF REVISION OF THE TEXT

October 2011

Information for Malaysia

PRESCRIPTION DRUG

Reg. No. MAL20071631ACZ

Imported by: DKSH Malaysia Sdn Bhd,

B-11-01, The Ascent, Paradigm,

No.1, Jalan SS7/26A, Kelana Jaya,

47301 Petaling Jaya, Selangor Darul Ehsan

Manufactured by:	for:  medinova SWITZERLAND
Haupt Pharma Amareg GmbH Donaustauer Str. 378 93 055 Regensburg Germany	Medinova AG Switzerland

806645-51738076-05/18